

displacement = $2r = 2 \times 0.50 \text{ m} = 1.0 \text{ m}$.

10

C

Let m be the mass of the satellite and M be the mass of Earth.

$$\text{GPE of orbiting satellite} = -\frac{GMm}{r} \text{ and KE} = \frac{GMm}{2r}$$

When satellite moves closer to the surface of Earth, r decrease. From the equations, GPE becomes more negative, so GPE decreases. KE increases, hence orbital speed increases.

11

B